SANTHEEPKUMAR VEERASAMY

PRINCIPAL SOFTWARE ENGINEER | ENTERPRISE SYSTEMS ARCHITECT & TECHNICAL LEADER

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OVERVIEW

I'm a self-taught, obsessive software builder with 7+ years of experience taking complex systems from a blank repository to production cloud infrastructure. My journey started in a government school in Tamil Nadu on a 2GB RAM state-issued laptop, teaching myself programming through documentation and YouTube—a drive that took me from converting a college internship into a full-time role in under 3 months (Skematix), to fast-tracking from Senior Engineer to Principal Software Engineer (HOD Digitals), leading tier-1 enterprise deliveries at Heady (shipping a Chuck E. Cheese web MVP solo in 2 months vs. an 8-month estimate), and now owning end-to-end architecture as a Staff Engineer at Tuna Software Solution and Lead Software Engineer (Part-Time Contractor) at Science & Humans.

As a Principal-level architect and hands-on builder, I own the entire technical lifecycle: designing distributed microservices & inter-service communication layers, engineering authentication/RBAC and multi-channel notification engines from the ground up, orchestrating Kubernetes (K8s) and AWS infrastructure with automated CI/CD, and mentoring engineering teams to ship with high velocity (+35%). Outside of client work, I constantly build and ship my own production platforms—including StrawHatDevs (strawhatdevs.com + 200+ architecture and programming videos on YouTube), ChimeFlow (a multi-tenant omnichannel notification & cadence platform), ENVault (a zero-knowledge encrypted .env sync platform with a native Go CLI), and IncidentLens AI (an evidence-grounded incident triage & Runbook RAG engine).

CORE TECHNICAL COMPETENCIES

Cloud, K8s & DevOps:	Kubernetes (K8s), Docker, AWS (ECS, Lambda, S3, RDS, CloudFront, IAM), GCP, GitHub Actions, Bitbucket Pipelines, Jenkins, Vercel, Terraform / IaC, Sentry, Axiom
Architecture & Systems:	Microservices, Inter-Service Communication, Event-Driven Systems, Clean/Hexagonal Architecture, Monorepos (Turborepo), Auth/RBAC, Notification Engines, Redis, Kafka, RabbitMQ
Frontend Engineering:	React, Next.js (App & Pages Router), TypeScript, JavaScript (ES6+), SvelteKit / Svelte 5, React Native, Expo, Tailwind CSS, State Management (Zustand, Redux)
Backend & Languages:	Node.js, NestJS, Go (Golang), Bun, Express.js, RESTful APIs, GraphQL, WebCrypto / AES-256-GCM Encryption, IoT Hardware & POS Peripherals
Databases & Storage:	PostgreSQL (Raw SQL, RLS, FTS/tsvector, SKIP LOCKED), MongoDB, Supabase, Drizzle ORM, TypeORM, Prisma, Sequelize, Mongoose, Query Tuning
Testing, AI & Quality:	Playwright, Vitest, Jest, React Testing Library, LLM RAG & Bounded Agents, Automated Eval Harnesses, CI/CD Quality Gates

PROFESSIONAL EXPERIENCE

Staff Engineer | Tuna Software Solution
Jun 2025 – Present

- **End-to-End Architecture & Delivery:** Owned full-stack architecture, cloud infrastructure, backend microservices, and frontend platforms across two concurrent enterprise initiatives, driving feature releases on schedule.
- **Pony Up IoT Kiosk & Cloud Web Migration:** Architected and delivered an Android self-service kiosk (React Native/Expo) integrating physical IoT peripherals (payment terminals, receipt printers, cash drawers), then led its rebuild as a cloud web platform to eliminate deployment friction and improve cross-device compatibility.
- **CI/CD Automation & Engineering Standards:** Established structured code review processes, automated CI/CD pipelines (GitHub Actions/Docker), and coding standards—boosting team delivery velocity by 35% and reducing production defect escapes.
- **Stakeholder & Product Alignment:** Collaborated directly with product stakeholders to translate business requirements into scalable technical solutions, acting as the primary technical point of contact.

Lead Software Engineer (Part-Time Contractor) | Science & Humans
Jun 2025 – Present

- **Ground-Up Ownership & Product Planning:** Contributed from the ground up with full end-to-end ownership—partnering directly with Product Managers to plan implementations, estimate modules, architect solutions, and ship production releases.
- **Authentication, Authorization & Notification Engine:** Architected and built the entire authentication and role-based authorization (RBAC) system alongside a centralized multi-channel notification system for the platform.
- **Microservices & Communication Layer:** Set up decoupled backend microservices and engineered the inter-service communication layer from scratch while resolving critical system bottlenecks and bugs to harden platform stability.
- **Kubernetes (K8s) & DevOps Configuration:** Configured and managed containerized application deployments on Kubernetes (K8s) and built automated CI/CD DevOps workflows for reliable, zero-downtime releases.
- **Mentoring & Team Ramp-Up:** Guided, onboarded, and mentored junior developers joining the project, conducting code reviews and unblocking engineering execution across modules.

Senior Software Engineer | Heady
May 2023 – Jun 2025

- **Enterprise Client Leadership:** Served as primary technical lead interfacing with tier-1 enterprise clients (HeyDay Skin Care, Chuck E. Cheese, The Lash Lounge), coordinating with PMs and stakeholders to define scope, set milestones, and ship production systems.
- **Chuck E. Cheese Web MVP (75% Faster Delivery):** Delivered the Chuck E. Cheese web kiosk MVP in 2 months as a solo contributor, compressing an 8-month original project timeline by 75% through focused full-stack engineering and rapid iteration.
- **HeyDay Skin Care Microservices Ecosystem:** Architected a microservices-based backend integration with Boulevard (booking platform), Redis caching, Sentry error monitoring, and AWS infrastructure sustaining 99.9% reliability.
- **AI-Assisted Workflows & Mentorship:** Mentored 6+ junior and mid-level developers on modern development practices, AI-assisted coding workflows, and architectural patterns that improved code quality and reduced regression rates.

Principal Software Engineer (Promoted from Senior Frontend Engineer) | HOD Digitals
Sep 2021 – May 2023

- **Fast-Track Promotion (Senior Engineer → Principal Engineer):** Joined as Senior Frontend Engineer contributing to Project Gen and concurrent client systems; promoted to Principal Software Engineer (Nov 2022) to lead technical direction and architecture across multiple product lines.
- **E-Commerce & Scalable System Architecture:** Architected and shipped full-stack e-commerce platforms (Tamil Mart web & iOS/Android mobile apps with Next.js, Magento GraphQL, React Native), CRM subscription systems (Vmeals on AWS), and Playwright CI/CD test automation (AMS).
- **Cross-Functional Leadership & Mentorship:** Oversaw project planning, system design, and cloud infrastructure decisions while acting as primary technical liaison between engineering teams and business stakeholders.

Full Stack Developer (Converted from Intern in <3 Months) | Skematix Technologies — Dec 2019 – Sep 2021

- **Rapid Intern-to-Full-Time Conversion:** Joined during college in 2019 as an Intern Software Engineer and earned conversion to full-time Full Stack Developer within 2–3 months due to rapid execution and deep ownership.
- **End-to-End Web Application Delivery:** Developed, optimized, and deployed multiple client-facing web applications across React, Node.js, Express, and relational/NoSQL databases from initial schema design to production hosting.

SELF-BUILT PRODUCTS, PLATFORMS & DEVELOPER COMMUNITY

ChimeFlow — Multi-Tenant Omnichannel Notification & Cadence Platform

Engineered a multi-tenant notification & cadence orchestration platform supporting Email (SendGrid), SMS (Twilio), and Push (Firebase FCM) using Clean/Hexagonal Architecture. Built a write-ahead delivery ledger, fair-claim background worker queues, idempotency keys, per-tenant rate limiting, suppression/consent engines, encrypted credentials at rest, and Razorpay billing.

Bun, TypeScript, SvelteKit / Svelte 5, PostgreSQL 17, Clerk Auth, SendGrid, Twilio, Firebase Admin, Docker, AWS

ENVault — Zero-Knowledge Encrypted .env Sync Platform & Native Go CLI

Built an end-to-end client-encrypted team environment variable & secret synchronization platform with browser WebCrypto key derivation, atomic revision rollback, and a native cross-platform Go CLI (github.com/Santheepkumar/envault-cli) integrated with macOS Keychain, Windows Credential Manager, and Linux Secret Service.

Go (Golang CLI), Next.js, TypeScript, PostgreSQL 17, Drizzle ORM, WebCrypto API, Better Auth, Docker

IncidentLens AI — Evidence-Grounded Incident Triage & Runbook RAG Engine

Designed a multi-tenant AI production incident investigator connecting Sentry telemetry with Markdown runbooks (PostgreSQL tsvector + GIN Hybrid FTS RAG), AES-256-GCM secret vault, a bounded 6-step agent loop with deterministic [ev_xxx] citation verification, FOR UPDATE SKIP LOCKED job queues, and a 35-case AI evaluation harness (93.3% root-cause accuracy).

TypeScript, Turborepo, Node.js, PostgreSQL (Raw SQL + RLS), Auth0 OAuth2 PKCE, LLM Tool-Calling & RAG, Vitest

StrawHatDevs Platform (strawhatdevs.com) & Route-Atlas Open-Source Tooling

Architected strawhatdevs.com full-stack learning platform (200+ architecture/programming videos) alongside developer tools including Route-Atlas (published npm visual atlas for Next.js App Router) and SelectCheck (Tauri v2 + Cloudflare Workers AI assistant).

Next.js 15, React 19, TypeScript, Supabase, PostgreSQL, Sanity CMS, Tauri v2, Cloudflare Workers, Vercel CI/CD

KEY ENTERPRISE PROJECTS & PRODUCTION DEPLOYMENTS

Science & Humans — Healthcare Platform & Microservices Ecosystem

Built core platform architecture from the ground up with full end-to-end ownership: engineered the complete authentication & RBAC authorization system, event-driven notification engine, decoupled microservices & inter-service communication layer, and production Kubernetes (K8s) + CI/CD DevOps infrastructure while mentoring junior developers.

TypeScript, Node.js, Microservices, Auth/RBAC, Notification System, Kubernetes (K8s), Docker, CI/CD

Pony Up — Android Self-Service IoT Retail Kiosk & Cloud Web Platform

Engineered a React Native/Expo self-service kiosk deployed in shopping malls integrating physical IoT hardware (payment terminals, thermal receipt printers, cash drawers), and subsequently led its strategic rebuild into a cloud-hosted web platform.

React Native, Expo, Next.js, IoT Peripherals, Android Kiosk Mode, REST APIs, Docker, GitHub Actions

Chuck E. Cheese — Customer Experience Web Platform (2-Month Solo MVP)

Rebuilt the Chuck E. Cheese kiosk into a mobile-responsive customer experience web platform enabling instant product interaction via QR code scanning—delivered the production MVP solo in 2 months vs. an 8-month original estimate (75% faster time-to-market).

React, Next.js, TypeScript, Tailwind CSS, QR Integration, AWS CloudFront, Automated CI/CD

HeyDay Skin Care & The Lash Lounge — Enterprise Booking & Microservices

Architected high-concurrency appointment booking widgets, e-commerce shop apps, and microservices backends integrated with Boulevard API, Redis caching, Sentry monitoring, and AWS infrastructure sustaining 99.9% uptime.

React, Next.js, Node.js, NestJS, Redis, Microservices, AWS, Vercel, Bitbucket Pipelines, Sentry, Boulevard API

EDUCATION

Bachelor of Engineering (B.E.) — Computer Science & Engineering • Anna University, Tamil Nadu, India 2015 – 2019